

Comparative Analysis of BRCA1 Protein Sequences in Human, Mouse, and Rat Using Clustal Omega

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Independent Learning Project

Abstract

BRCA1 is an important tumor suppressor protein involved in DNA repair, genomic stability, and cell cycle regulation. In this study, a comparative bioinformatics analysis of BRCA1 protein sequences from human (*Homo sapiens*), mouse (*Mus musculus*), and rat (*Rattus norvegicus*) was performed using Clustal Omega. Protein sequences were retrieved from the UniProt database in FASTA format and subjected to Multiple Sequence Alignment (MSA) to identify conserved and variable regions among the three species. The alignment results revealed a high degree of sequence conservation, indicating the evolutionary and functional importance of the BRCA1 protein across mammals. Guide tree and phylogenetic tree analyses demonstrated that mouse and rat BRCA1 proteins are more closely related to each other than to human BRCA1, consistent with known evolutionary relationships. Visualization of aligned sequences further highlighted conserved amino acid residues essential for protein function. This study demonstrates the application of bioinformatics tools in comparative protein analysis and provides insight into the evolutionary conservation of the BRCA1 protein.

Introduction

Bioinformatics tools and databases provide valuable information about protein sequences, structure, function, and evolutionary relationships. Comparative sequence analysis is an important approach used to study conserved regions,

functional similarities, and evolutionary divergence among proteins from different organisms.

Multiple Sequence Alignment (MSA) is a widely used bioinformatics technique that aligns protein or nucleotide sequences to identify conserved amino acid residues and regions of similarity. Conserved regions often indicate structural or functional importance and help in understanding evolutionary relationships among species.

Clustal Omega is a powerful and commonly used tool for performing multiple sequence alignment and phylogenetic analysis. It generates sequence alignments, guide trees, and phylogenetic trees that help visualize similarities and evolutionary distances between proteins.

The BRCA1 protein is an important tumor suppressor involved in DNA repair, maintenance of genomic stability, and regulation of the cell cycle. Mutations in the BRCA1 gene are strongly associated with hereditary breast and ovarian cancers. Studying the conservation of BRCA1 across species helps in understanding its biological significance and evolutionary importance.

This study focuses on the comparative analysis of BRCA1 protein sequences from human (*Homo sapiens*), mouse (*Mus musculus*), and rat (*Rattus norvegicus*) using Clustal Omega. Multiple sequence alignment and phylogenetic analysis were performed to identify conserved regions and evolutionary relationships among the selected mammalian species.

Learning Objective

The objective of this study was to gain practical experience in performing comparative protein sequence analysis using bioinformatics databases and multiple sequence alignment tools. This activity helped in understanding the evolutionary conservation, sequence similarity, and functional significance of the

BRCA1 protein across different mammalian species. The study also provided hands-on experience in retrieving protein sequences from biological databases, performing sequence alignment using Clustal Omega, and interpreting phylogenetic relationships among organisms.

Materials and Methods

Protein data used in this study was obtained from the UniProt database. The proteins selected for comparative analysis were:

- Protein: Breast cancer type 1 susceptibility protein (BRCA1)
- Organisms: *Homo sapiens*, *Mus musculus*, and *Rattus norvegicus*
- Database: UniProt
- Tool Used: Clustal Omega

Workflow of the Analysis

1. Search for BRCA1 protein sequences in the UniProt database
2. Select reviewed BRCA1 entries for human, mouse, and rat
3. Retrieve protein sequences in FASTA format
4. Combine the sequences into a single FASTA file
5. Perform Multiple Sequence Alignment using Clustal Omega
6. Analyze conserved regions and phylogenetic relationships
7. Document observations and screenshots

Results

The Multiple Sequence Alignment of BRCA1 proteins from human, mouse, and rat using Clustal Omega revealed a high degree of sequence similarity among the selected mammalian species. Several conserved amino acid residues were identified throughout the alignment, indicating the evolutionary and functional importance of the BRCA1 protein.

The alignment output displayed multiple conserved regions marked by alignment symbols such as asterisks (*), which represent fully conserved amino acid residues across all sequences. Additional similarity symbols such as colons (:) and periods (.) indicated strongly and weakly conserved substitutions among the aligned sequences.

The phylogenetic and guide tree analyses demonstrated evolutionary relationships among the selected organisms. The tree showed that mouse and rat BRCA1 proteins are more closely related to each other than to human BRCA1, consistent with known mammalian evolutionary relationships.

Visualization of the alignment through color-coded sequence representation further highlighted conserved domains and sequence similarities among the three species. These results suggest that important functional regions of the BRCA1 protein have remained highly conserved during evolution.

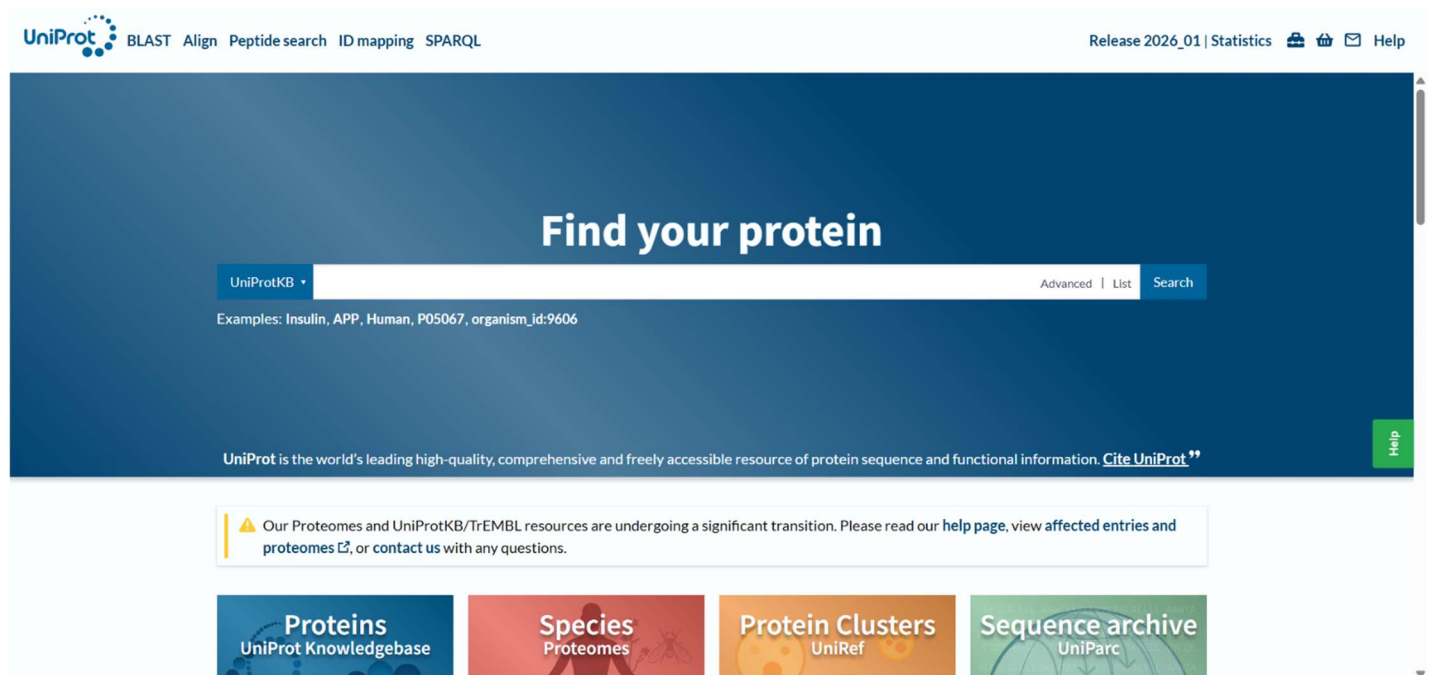


Figure 1: Homepage of the UniProt database showing navigation tools and access to protein sequence and functional information.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB brca1 human Advanced List Search Help					
Status Reviewed (Swiss-Prot) (196) Unreviewed (TrEMBL) (1,257) Popular organisms Human (673) Mouse (25) Zebrafish (23) A. thaliana (1) C. elegans (1) Taxonomy Filter by taxonomy Group by Taxonomy Keywords Gene Ontology Enzyme Class UniProtKB 1,453 results Tools Download (1k) Add View: Cards Table Customize columns Share					
Entry	Entry Name	Protein Names	Gene Names	Organism	Length
☐ P38398	BRCA1_HUMAN	Breast cancer type 1 susceptibility protein[...]	BRCA1, RNF53	Homo sapiens (Human)	1,863 AA
☐ Q95153	BRCA1_CANLF	Breast cancer type 1 susceptibility protein homolog[...]	BRCA1	Canis lupus familiaris (Dog) (Canis familiaris)	1,878 AA
☐ Q8RXD4	BRCA1_ARATH	Protein BREAST CANCER SUSCEPTIBILITY 1 homolog[...]	BRCA1, At4g21070, T13K14.230	Arabidopsis thaliana (Mouse-ear cress)	941 AA
☐ Q9BX63	FANCI_HUMAN	Fanconi anemia group J protein[...]	BRIP1, BACH1, FANCI	Homo sapiens (Human)	1,249 AA
☐ P48754	BRCA1_MOUSE	Breast cancer type 1 susceptibility protein homolog[...]	Brca1	Mus musculus (Mouse)	1,812 AA
☐ Q9GKK8	BRCA1_PANTR	Breast cancer type 1 susceptibility protein homolog[...]	BRCA1	Pan troglodytes (Chimpanzee)	1,863 AA
☐ Q8WX92	NELFB_HUMAN	Negative elongation factor B[...]	NELFB, COBRA1, KIAA1182	Homo sapiens (Human)	580 AA
☐ Q6PJG6	BRAT1_HUMAN	Integrator complex assembly factor BRAT1[...]	BRAT1, BAAT1, C7orf27	Homo sapiens (Human)	821 AA
☐ Q99728	BARD1_HUMAN	BRCA1-associated RING domain	BARD1	Homo sapiens (Human)	777 AA

Figure 2: Search results for BRCA1 protein sequences in the UniProt database showing entries from different organisms.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced List Search Help					
Function Names & Taxonomy Subcellular Location Disease & Variants PTM/Processing Expression Interaction Structure Family & Domains Sequence & Isoforms Similar Proteins Homologs P38398 · BRCA1_HUMAN Proteinⁱ Breast cancer type 1 susceptibility protein Amino acids 1863 (go to sequence) Geneⁱ BRCA1 Protein existenceⁱ Evidence at protein level Statusⁱ UniProtKB reviewed (Swiss-Prot) Annotation scoreⁱ 5/5 Organismⁱ Homo sapiens (Human)					
Entry Variant viewer Feature viewer Genomic coordinates Publications External links History Tools Download Add Community curated (1) Add a publication Entry feedback					
Functionⁱ E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage (PubMed:10500182, PubMed:12887909, PubMed:12890688, PubMed:14976165, PubMed:16818604, PubMed:17525340, PubMed:19261748). It is unclear whether it also mediates the formation of other types of polyubiquitin chains (PubMed:12890688). The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability (PubMed:12890688, PubMed:14976165, PubMed:20351172). Regulates centrosomal microtubule nucleation (PubMed:18056443). Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle (PubMed:10724175, PubMed:11836499, PubMed:12183412, PubMed:19261748). Required for FANCD2 targeting to sites of DNA damage (PubMed:12887909).					

Figure 3: Reviewed UniProt entry page for the human BRCA1 protein displaying protein information and annotations.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

Function Entry Variant viewer **0,952** Feature viewer Genomic coordinates Publications External links History

Sequence & Isoformsⁱ

Align isoforms (8) Add isoforms

Sequence statusⁱ Complete

This entry describes 8 isoformsⁱ produced by **Alternative splicing & Alternative initiation**.

P38398-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name 1 See also sequence in UniParc or sequence clusters in UniRef

Tools Download Add Highlight Copy sequence

Length 1,863 Last updated 1995-02-01 v2
Mass (Da) 207,721 MD5 Checksumⁱ E40F752DED675E2F7C99142EBB2607A

MDLSALRVEEVQNVINAMQKILECPICLELIEKEPVSTKCDHIFCKFCMLKLLNQKKGPSQ
CPLCKNDITKRSIQESTRFSQLVEELLKIIICAFQLDTGLEIYANSYNFAKKENNSPEHLKD
EVSIIQSMGYRNRRAKRLQLQSEPNPSLQETSLSVQLSNLGTVRTLRKQRIQPKTSVYI
ELGSDSSEDTVNKATYCSVGDQELLQITPQGRDEISLDSAKKAACEFSETDVTNTEHHQ
PSNNDLNTTEKRAAERHPEKYQGSSVSNLHVPCGTNTHASSLQHENSLLLTKDRMNVE
KAEFCNKSKQPGGLARSQHNRWAGSKETCNDRRPTSTKVKDLNADPLCKERKEWNKQKLP
SENPRDTEVPWITLNSIIQKVNWFSSRDELLGSDSDHGESESNKAVADVLDVLEVD
EYSGSSEKIDLLASDPHEALICKSERVHKSSESNIEDKIFGKTYRKKASLPNLSHVTEN
LIIQAFVTEPQIIQERPLTNKLRKRRTSGLHPEDFIKKADLAVQKTPMINQGTNQT
QNGQVMNITNSGHENKTGKDSIQNEKNPNPIESLEKESAFKTKAEPISSSISNMELELNI
HNSKAPKKNRLRRKSSSTRHIALELVVSRNLSPPNCTELQIDSCSSSEIEKKKKYNQMPV
RHSRNLQLMGKEPATGAKSKNKPNEQTSKRHDSDTFPELKL TNAPGSFTKCSNTSELKE
FVNPSLPREEKEEKLTVKVSNNAEADPKDMLSGERVLTQERSVESSISLVPDGYGTQ
ESTSLLEVSTLGAKTEPNKCVSQCAAFENPKGLIHGCSKDNRRNDETFKYPGLGHEVNHS
RETSIEMEESLDAQYLQNTFKVSKRQSFAPSNPGNAEEECATFSAHSGSLKKQSPKVT
FECEQKEENQGNESNIKPQVTNITAGFPVVGQDKPVDNAKCSIKGGSRFCLSSQFRG
NETGLITPNKHGLLQNPYRIPPLFPIKSFVKTKCKKNLLEENFEHSMSPEREMGNENIP
STVSTISRNNIRENVKEASSNINEVGSSSTNEVGSSSINEIGSSDENIQAEIGRNRGPKL
NAMLRLGLVQPEVYKQSLPGSNCKHPEIKKQYEEVVQTVNTDFSPYLISDNLEQPMGSS
HASQVCSETPDDLDDGEIKEDTSAENDIKESSAVFSKSVQKGLSRSPSPFTHTHLAQ
GYRRGAKKLESSEENLSEDEELPCFQHLLFGKVNIPISQSTRHSTVATECLSKNTEENL
LSLKNSLNDSCNQVILAKASQEHHLSEETKCSASLFSQCSLEDLTANTNTQDPFLIGS
SKQMRHQSESQGVGLSDKELVSDDEERGTLLENNQEEQSMDSNLGEAASGCSETSVSE
DCSGLSQSDILTTQQRDTMQHNLIKLQEMAELEAVLEQHGSGPSNSYPSIISDSSALE
DLRNPEQSTSEKAVLTSQKSSEYPTSQNPEGLSADKFEVSADSSSTSKNKEPGVERSSPSK
CPSLDDRWMHSCSGSLQNRNYPQEEELIKVVDVEEQLEESGPHDLTETSYLPRQDLEG
TPYLESGISLFSDDPESDPSEDRAPEARSVGNIPSSSALKVPQLKVAESAQSPAAAHIT
DTAGYNAMESVSREKPELTASTERVNRKMSMVVSGLTPEEFMLVYKFARKKHITLTNLI
TEETTHVVMKDAEFVCERTLKYLFGIAGGKVVVSYFWVTQSTIKERKMLNEHDFEVRG
VNGRNGHQPKRARESQRKIFRGLTCCYGPFTNMPTDQLEWVQLCGASVVKELSSFTL
GTGVHPIVVVQPDADWEDNGFHAIGQMCEAPVVTREWLDSVALYQCQELDTYLIPQIPH
SHY

Figure 4: Sequence & Isoforms section of the BRCA1 UniProt entry showing amino acid sequence information.

```
>sp|P38398|BRCA1_HUMAN Breast cancer type 1 susceptibility protein OS=Homo sapiens OX=9606 GN=BRCA1 PE=1 SV=2
MDLSALRVEEVQNVINAMQKILECPICLELIEKEPVSTKCDHIFCKFCMLKLLNQKKGPSQ
CPLCKNDITKRSIQESTRFSQLVEELLKIIICAFQLDTGLEIYANSYNFAKKENNSPEHLKD
EVSIIQSMGYRNRRAKRLQLQSEPNPSLQETSLSVQLSNLGTVRTLRKQRIQPKTSVYI
ELGSDSSEDTVNKATYCSVGDQELLQITPQGRDEISLDSAKKAACEFSETDVTNTEHHQ
PSNNDLNTTEKRAAERHPEKYQGSSVSNLHVPCGTNTHASSLQHENSLLLTKDRMNVE
KAEFCNKSKQPGGLARSQHNRWAGSKETCNDRRPTSTKVKDLNADPLCKERKEWNKQKLP
SENPRDTEVPWITLNSIIQKVNWFSSRDELLGSDSDHGESESNKAVADVLDVLEVD
EYSGSSEKIDLLASDPHEALICKSERVHKSSESNIEDKIFGKTYRKKASLPNLSHVTEN
LIIQAFVTEPQIIQERPLTNKLRKRRTSGLHPEDFIKKADLAVQKTPMINQGTNQT
QNGQVMNITNSGHENKTGKDSIQNEKNPNPIESLEKESAFKTKAEPISSSISNMELELNI
HNSKAPKKNRLRRKSSSTRHIALELVVSRNLSPPNCTELQIDSCSSSEIEKKKKYNQMPV
RHSRNLQLMGKEPATGAKSKNKPNEQTSKRHDSDTFPELKL TNAPGSFTKCSNTSELKE
FVNPSLPREEKEEKLTVKVSNNAEADPKDMLSGERVLTQERSVESSISLVPDGYGTQ
ESTSLLEVSTLGAKTEPNKCVSQCAAFENPKGLIHGCSKDNRRNDETFKYPGLGHEVNHS
RETSIEMEESLDAQYLQNTFKVSKRQSFAPSNPGNAEEECATFSAHSGSLKKQSPKVT
FECEQKEENQGNESNIKPQVTNITAGFPVVGQDKPVDNAKCSIKGGSRFCLSSQFRG
NETGLITPNKHGLLQNPYRIPPLFPIKSFVKTKCKKNLLEENFEHSMSPEREMGNENIP
STVSTISRNNIRENVKEASSNINEVGSSSTNEVGSSSINEIGSSDENIQAEIGRNRGPKL
NAMLRLGLVQPEVYKQSLPGSNCKHPEIKKQYEEVVQTVNTDFSPYLISDNLEQPMGSS
HASQVCSETPDDLDDGEIKEDTSAENDIKESSAVFSKSVQKGLSRSPSPFTHTHLAQ
GYRRGAKKLESSEENLSEDEELPCFQHLLFGKVNIPISQSTRHSTVATECLSKNTEENL
LSLKNSLNDSCNQVILAKASQEHHLSEETKCSASLFSQCSLEDLTANTNTQDPFLIGS
SKQMRHQSESQGVGLSDKELVSDDEERGTLLENNQEEQSMDSNLGEAASGCSETSVSE
DCSGLSQSDILTTQQRDTMQHNLIKLQEMAELEAVLEQHGSGPSNSYPSIISDSSALE
DLRNPEQSTSEKAVLTSQKSSEYPTSQNPEGLSADKFEVSADSSSTSKNKEPGVERSSPSK
CPSLDDRWMHSCSGSLQNRNYPQEEELIKVVDVEEQLEESGPHDLTETSYLPRQDLEG
TPYLESGISLFSDDPESDPSEDRAPEARSVGNIPSSSALKVPQLKVAESAQSPAAAHIT
DTAGYNAMESVSREKPELTASTERVNRKMSMVVSGLTPEEFMLVYKFARKKHITLTNLI
TEETTHVVMKDAEFVCERTLKYLFGIAGGKVVVSYFWVTQSTIKERKMLNEHDFEVRG
VNGRNGHQPKRARESQRKIFRGLTCCYGPFTNMPTDQLEWVQLCGASVVKELSSFTL
GTGVHPIVVVQPDADWEDNGFHAIGQMCEAPVVTREWLDSVALYQCQELDTYLIPQIPH
SHY
```

Figure 5: FASTA format sequence of the human BRCA1 protein retrieved from the UniProt database.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

P48754 · BRCA1_MOUSE

Names & Taxonomy	Protein ⁱ Breast cancer type 1 susceptibility protein homolog	Amino acids	1812 (go to sequence)
Subcellular Location	Gene ⁱ Brca1	Protein existence ⁱ	Evidence at protein level
Phenotypes & Variants	Status ⁱ UniProtKB reviewed (Swiss-Prot)	Annotation score ⁱ	5/5
PTM/Processing	Organism ⁱ Mus musculus (Mouse)		

Expression Interaction Structure Tools Download Add Add a publication Entry feedback

Entry Variant viewer 148 Feature viewer Genomic coordinates Publications External links History

Functionⁱ

E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage. It is unclear whether it also mediates the formation of other types of polyubiquitin chains. The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability. Regulates centrosomal microtubule nucleation. Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle. Required for FANCD2 targeting to sites of DNA damage. Inhibits lipid synthesis by binding to inactive phosphorylated ACACA and preventing its dephosphorylation. Contributes to homologous recombination repair (HRR) via its direct interaction with PALB2, fine-tunes recombinational repair partly through its modulatory role in the PALB2-dependent loading of BRCA2-RAD51 repair machinery at DNA breaks. Component of the BRCA1-RBBP8 complex which regulates CHEK1 activation and controls cell cycle G2/M checkpoints on DNA damage via BRCA1-mediated ubiquitination of RBBP8. Acts as a transcriptional activator. [By Similarity](#)

Catalytic activityⁱ

Figure 6: Reviewed UniProt entry page for the mouse BRCA1 protein.

UniProt BLAST Align Peptide search ID mapping SPARQL UniProtKB Advanced | List Search

O54952 · BRCA1_RAT

Names & Taxonomy	Protein ⁱ Breast cancer type 1 susceptibility protein homolog	Amino acids	1817 (go to sequence)
Subcellular Location	Gene ⁱ Brca1	Protein existence ⁱ	Evidence at protein level
Phenotypes & Variants	Status ⁱ UniProtKB reviewed (Swiss-Prot)	Annotation score ⁱ	5/5
PTM/Processing	Organism ⁱ Rattus norvegicus (Rat)		

Expression Interaction Structure Tools Download Add Add a publication Entry feedback

Entry Variant viewer Feature viewer Genomic coordinates Publications External links History

Functionⁱ

E3 ubiquitin-protein ligase that specifically mediates the formation of 'Lys-6'-linked polyubiquitin chains and plays a central role in DNA repair by facilitating cellular responses to DNA damage. It is unclear whether it also mediates the formation of other types of polyubiquitin chains. The BRCA1-BARD1 heterodimer coordinates a diverse range of cellular pathways such as DNA damage repair, ubiquitination and transcriptional regulation to maintain genomic stability. Regulates centrosomal microtubule nucleation. Required for appropriate cell cycle arrests after ionizing irradiation in both the S-phase and the G2 phase of the cell cycle. Required for FANCD2 targeting to sites of DNA damage. Inhibits lipid synthesis by binding to inactive phosphorylated ACACA and preventing its dephosphorylation. Contributes to homologous recombination repair (HRR) via its direct interaction with PALB2, fine-tunes recombinational repair partly through its modulatory role in the PALB2-dependent loading of BRCA2-RAD51 repair machinery at DNA breaks. Component of the BRCA1-RBBP8 complex which regulates CHEK1 activation and controls cell cycle G2/M checkpoints on DNA damage via BRCA1-mediated ubiquitination of RBBP8. Acts as a transcriptional activator. [By Similarity](#)

Catalytic activityⁱ

Figure 7: Reviewed UniProt entry page for the rat BRCA1 protein.

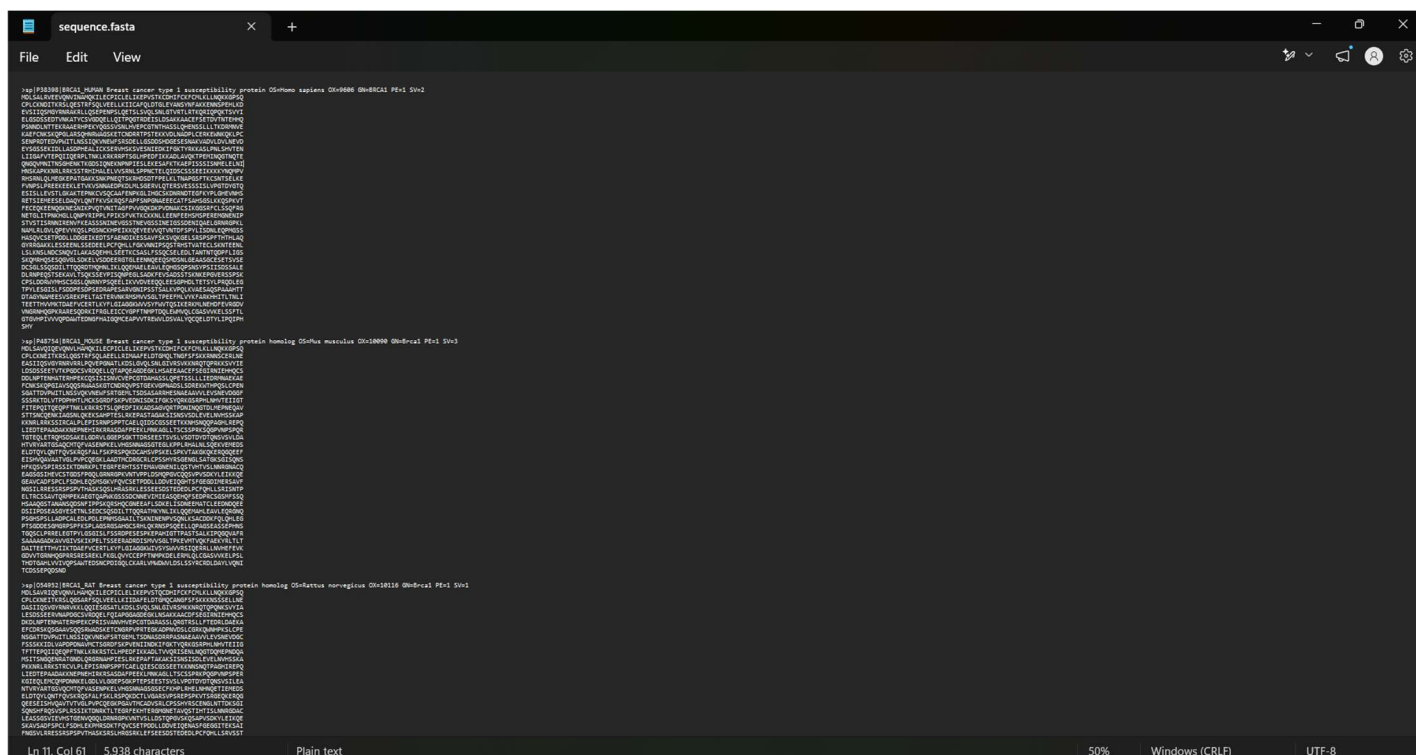


Figure 8: Combined FASTA sequences of human, mouse, and rat BRCA1 proteins prepared for Multiple Sequence Alignment analysis.

Paste your sequence here - or use the example sequence

Choose File

sequence.fasta

Use the example

Clear sequence

More example inputs

Parameters

OUTPUT FORMAT ⓘ

ClustalW with character counts

▼

More options ▼

Submit

Title

Clustal Omega's job

Submit

Figure 9: Clustal Omega input page showing uploaded BRCA1 protein sequences before alignment submission.

Download

sp P38398 BRCA1_HUMAN	MDLSAVRVEEQVNVHAMQKILECPTCLLEIKEPVSTQCDDHIFCFMKLLKNQKKGPSQ	60
sp P48754 BRCA1_MOUSE	MDLSAVRIQEQQVNVHAMQKILECPTCLLEIKEPVSTQCDDHIFCFMKLLKNQKKGPSQ	60
sp O54952 BRCA1_RAT	MDLSAVRIQEQQVNVHAMQKILECPTCLLEIKEPVSTQCDDHIFCFMKLLKNQKKGPSQ	60
	*****:::*****:*****:*****:*****:*****:*****:*****	
sp P38398 BRCA1_HUMAN	CPLCKNDITKRSLQESTRFSQLVEELLKIICAFQLDGLLEYANSYNFAKKENSPEHLKD	120
sp P48754 BRCA1_MOUSE	CPLCKNEITKRSLQGSTRFSQLAEELLIRIMAAFELDTGMQLTGNGSFSSKNNNSCERLNE	120
sp O54952 BRCA1_RAT	CPLCKNEITKRSLQGSARFSQVLVEELLIIDAFELDTGMQNGSFCFKNNKSSELLENE	120
	*****:***** *:*****,**:*: *: ***** :*,.:*,*,*,* * **:	
sp P38398 BRCA1_HUMAN	EVSIIQSGMYGNRAKRLQSEPENPSLQETSLSVLQSLNLTGVRTLRTRKQRIQPQKTSVYI	180
sp P48754 BRCA1_MOUSE	EASIIQSGVGYNRVNLPPQVEGNATLK-DLSGLVQLSNLGIVRSVKNNKRTQRPKRSVYI	179
sp O54952 BRCA1_RAT	DASIIQSGVGYNRVKKLQETIGESGATLK-DSLVQLSNLGIVRSVKNNKRTQRPQNYSVI	179
	,:****:,*****,:** * , .*: **,,***** **:::,:**:,****	
sp P38398 BRCA1_HUMAN	ELGSDSSEDTVNKATYCSVGDAQELLQITPQGTDRDEISLDAKKAACEFSEDTVNTTEHHQ	240
sp P48754 BRCA1_MOUSE	ELDSDSSEETVPKGDSVRDQLQLTAPQAGDEGKLHSAEAEACFFSEG-IRNIEHHQ	238
sp O54952 BRCA1_RAT	ALEDSDSEERNVPAGDCSVRDQLQLTAPGAGDEGKLHSAKKAACDFSEG-IRNIEHHQ	238
	* ****: *, ** ** *****:* *: * **,*,*,:***** ** : * ****	
sp P38398 BRCA1_HUMAN	PSNNDLNTTEKRAAHERPEKYQGSVSMLHVEPCGTNHASSLQHENSLLLTKDRMNV	300
sp P48754 BRCA1_MOUSE	CSD-DLNPTENNATERHPKEQQTSTSMNCVCEPGTDHASSLQPETSLILLIEDRMNAE	297
sp O54952 BRCA1_RAT	CSDKDLNPNTENNATERHPKECPRTISVANVHVCEPGTDARASSLQRGTRSLFTEDRLDAE	298
	:*****:*****:*****:*****:*****:*****:*****:*****:*****:	
sp P38398 BRCA1_HUMAN	KAEFCNKSKQPGGLARSQHNRWAGSKETCNDRRTPTSTEKVLDNADPLCKERKENWQKLP	360
sp P48754 BRCA1_MOUSE	KAEFCNKSKQPGIAVSQGSRWAAASKGTCDNRQVPSTGEKGVNADSLDREKWTHWPQLC	357
sp O54952 BRCA1_RAT	KAEFCDRSQSGAAVSQGSRWADSKETCNGRPVPRTEGAADPNVDLCGRQGNWHPKSLC	358
	:*****:*** ** *,* *:*****:*****:*****:*****:*****:*****:	

Figure 12: Alignment conservation symbols showing fully conserved (*), strongly similar (:), and weakly similar (.) amino acid residues.

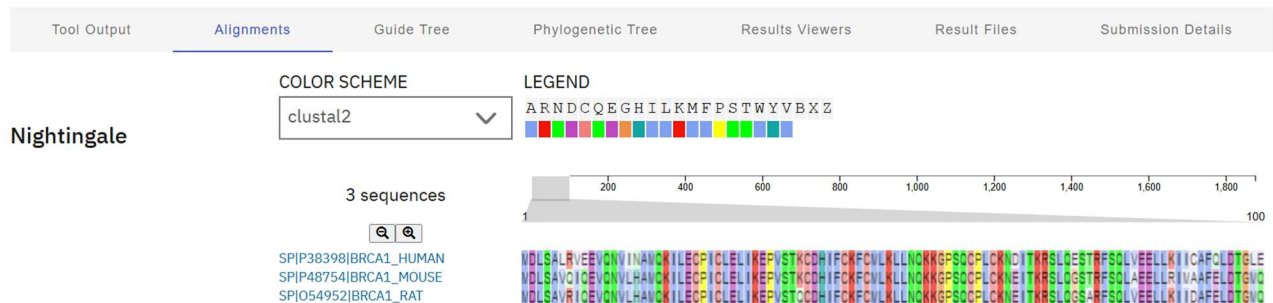


Figure 13: Nightingale alignment visualization representing sequence conservation across aligned BRCA1 proteins.

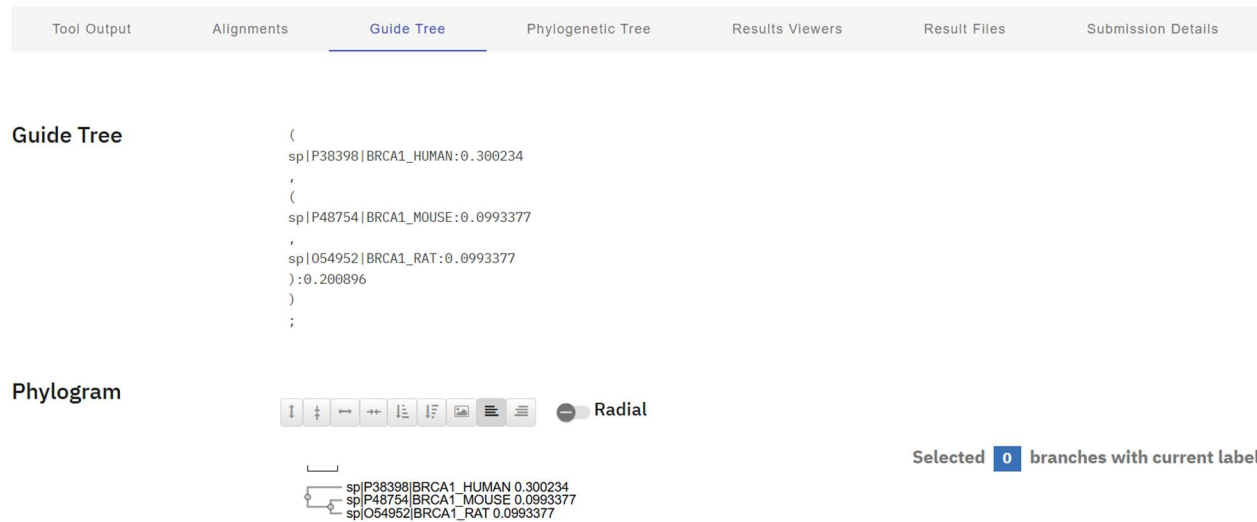


Figure 14: Guide tree generated from Multiple Sequence Alignment showing sequence clustering relationships among the selected species.



Figure 15: Phylogenetic tree illustrating the evolutionary relationships among human, mouse, and rat BRCA1 proteins.

Discussion

The results obtained from the Multiple Sequence Alignment of BRCA1 proteins demonstrate the importance of comparative bioinformatics analysis in understanding protein conservation and evolutionary relationships. The

alignment generated using Clustal Omega revealed several highly conserved amino acid residues among human, mouse, and rat BRCA1 proteins.

The presence of conserved regions suggests that the BRCA1 protein performs essential biological functions that have been maintained throughout mammalian evolution. Since BRCA1 plays a crucial role in DNA repair and maintenance of genomic stability, conservation of important residues is necessary for proper protein function.

The phylogenetic analysis showed that mouse and rat BRCA1 proteins are more closely related to each other than to human BRCA1, which is consistent with known evolutionary relationships among mammals. The alignment also showed only minor sequence variations, indicating a high degree of similarity among the selected species.

Overall, this study demonstrates the usefulness of Multiple Sequence Alignment tools in studying evolutionary conservation, sequence similarity, and functional significance of proteins using bioinformatics approaches.

Limitations of the Study

Although the comparative analysis of BRCA1 protein sequences provided useful insights into sequence conservation and evolutionary relationships, certain limitations are associated with the study.

Firstly, the analysis was entirely based on computational tools and publicly available database information without experimental validation. The accuracy of the results depends on the quality of the available sequence data.

Secondly, the study focused only on sequence alignment and phylogenetic analysis. Detailed structural analysis, mutation analysis, and protein interaction studies were not included.

Thirdly, the comparative analysis was limited to only three mammalian species. Including additional organisms could provide a broader understanding of BRCA1 evolutionary conservation across different species.

Despite these limitations, the study provides a clear understanding of the application of Multiple Sequence Alignment tools in comparative protein analysis and evolutionary studies.

Future Scope

The present study provides a basic understanding of comparative protein sequence analysis using Multiple Sequence Alignment tools. However, several advanced analyses can be carried out in the future to obtain deeper insights into the structure and function of the BRCA1 protein.

Future studies may include structural analysis of BRCA1 using databases such as the Protein Data Bank and structure prediction tools like AlphaFold. Visualization software such as PyMOL and Discovery Studio can also be used to study protein domains, binding sites, and molecular interactions.

Further research may involve mutation analysis, protein–protein interaction studies, and comparative analysis across a larger number of species to better understand the evolutionary conservation and biological significance of the BRCA1 protein.

Overall, advanced bioinformatics tools and computational approaches can provide a more comprehensive understanding of the structure, function, and evolutionary importance of BRCA1.

Conclusion

This study demonstrates the application of Multiple Sequence Alignment tools in comparative protein analysis. In this project, BRCA1 protein sequences from

human, mouse, and rat were analyzed using Clustal Omega to study sequence conservation and evolutionary relationships.

The alignment results revealed a high degree of similarity among the selected mammalian species, indicating that important regions of the BRCA1 protein are evolutionarily conserved. The phylogenetic analysis further showed close evolutionary relationships between the selected organisms.

This study provided practical experience in retrieving protein sequences from biological databases, performing sequence alignment, and interpreting phylogenetic relationships using bioinformatics tools. Overall, the project highlights the importance of computational approaches in understanding protein function, conservation, and biological significance.

References

- UniProt Consortium
- Clustal Omega – EMBL-EBI Multiple Sequence Alignment Tool
- National Center for Biotechnology Information (NCBI)
- Altschul SF et al. Basic Local Alignment Search Tool (BLAST)
- Mount DW. Bioinformatics: Sequence and Genome Analysis
- European Molecular Biology Laboratory – European Bioinformatics Institute (EMBL-EBI)